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Research paper

Explainable artificial intelligence framework for wind turbine fault detection using random forest – Extreme gradient boosting hybrid model

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ABSTRACT

Though wind energy has great promise for clean energy generation in India, operational inefficiencies and underutilization still present major obstacles. Although installed wind capacity exceeds 51. 3 GW, actual power generation is still significantly lower than predicted mostly because of weak fault detection and maintenance techniques. Existing machine learning (ML) methods offer high accuracy but typically lack transparency in their forecasts, therefore making it difficult for engineers to correctly interpret and act on model outputs. This research aims to develop an understandable and high-performance anomaly detection model using real-time SCADA data from an Indian wind plant. This research aims to develop an understandable and high-performance anomaly detection model using real-time SCADA data from an Indian wind plant. A hybrid ensemble approach integrating Random Forest and XGBoost is proposed, combined with Local Interpretable Model-Agnostic Explanations (LIME) to provide local interpretability of predictions. The model was trained and evaluated on actual SCADA data using SelectKBest for feature selection, SMOTE for handling class imbalance, and RandomizedSearchCV for hyperparameter optimization. The tuned hybrid model achieved outstanding performance, with an accuracy of 0.9995, F1-score of 0.9995, and minimal error rates (MAE and MSE = 0.00052). LIME-based interpretability highlighted key features driving predictions, with Nacelle Temperature and Gearbox Temperature consistently emerging as critical indicators of turbine braking events, underscoring the importance of thermal variables in fault diagnosis. The findings suggest that interpretable machine learning not only enhances root cause analysis but also supports proactive maintenance, particularly by emphasizing improvements to cooling systems to reduce thermal failures. By providing transparent and reliable insights, the proposed solution enables wind farm operators to make informed, timely decisions, thereby improving turbine reliability and energy yield. The framework is practical, explainable, and well-suited for deployment in smart wind farms, aligning with the United Nations Sustainable Development Goals, including SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 12 (Responsible Consumption and Production)

1. Introduction

Wind energy has become one of the most important renewable energy sources due to its sustainability, cost effectiveness, and role in reducing greenhouse gas emissions. The reliability of wind turbines remains a major issue, nevertheless, given their challenging operating dynamics and exposure to harsh environmental conditions. Unexpected problems in parts such as the gearbox, generator and blades can cause a lot of downtime and maintenance work, which lowers the efficiency of power generation. For raising the operational performance of wind turbines, therefore, fault detection and predictive maintenance techniques have become absolutely vital [1].

With 430 GW, China tops installed wind power capability as of mid-2025; the United States follows with 150 GW; Germany 68 GW; and India 51. 3 GW. India has problems in efficient wind energy generation even if it is ranked fourth. Wind contributed only 78. 2 billion units (GWh). (See Fig. 1.)

7% of the total electricity production between April 2024 and February 2025, therefore emphasizing over 80% under utilization of installed capacity. Low capacity utilization factor (CUF), poor maintenance, tur-

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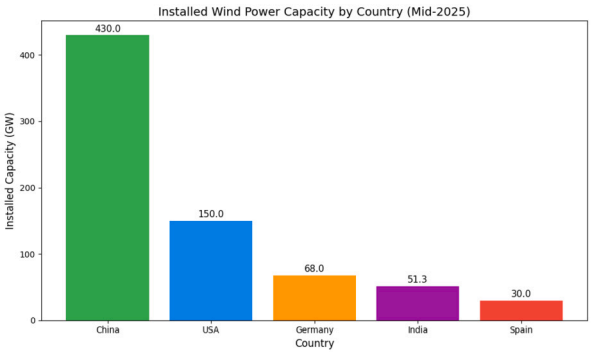


Fig. 1. Installed Wind power Capacity.

bine failures, transmission delays, and under utilization of local manufacturing capacity (over 18 GW annually) are among contributing elements. Although India's theoretical wind capacity exceeds 1100 GW (at 150 m hub height), various technological, infrastructural, and policy-level obstacles restrict actual deployment.

While Tamil Nadu is a leader in wind energy capacity, its development is hampered by aging turbines, land restrictions, and grid integration problems as well as other factors. These restrictions lower operating efficiency and stop the full use of the energy produced. Therefore, research and investment are needed to modernize infrastructure and develop strategies that can maximize Tamil Nadu's wind energy potential.

Conventional condition monitoring systems usually make use of physical models and threshold-based alerts. Although helpful, these techniques have restricted flexibility when processing enormous volumes of high-dimensional and noisy SCADA (Supervisory Control and Data Acquisition) data. Data-driven techniques including deep learning (DL) and machine learning (ML) have lately drawn more and more interest for fault diagnosis and abnormality detection in wind turbines. Early warnings before significant failures happen can be found by using these methods to find secret patterns in SCADA data [2].

Several research have shown the potential of DL and ML models to spot turbine irregularities. Using SCADA data, Kusiak and Verma [4], for example, used data-driven predictive models to spot anomalies earlier than conventional techniques would have done. High detection accuracy has been seen in sophisticated models like autoencoders, Convolutional Neural Network (CNN) and a Gated Recurrent Unit (GRU) frameworks, and xtreme Gradient Boosting (XGBoost) -based predictive maintenance systems [2–6]. Most models, however, operate as black boxes with poor interpretability, which makes it difficult for operators and engineers to grasp the justification behind fault alarms.

By giving interpretability for ML predictions,

XAI—explainable artificial intelligence—has evolved to meet these difficulties. Methods like SHAP (SHapley Additive Explanations) [5] and LIME (Local Interpretable Model Agnostic Explanations) [6] measure how much each feature, like temperature, vibration, or wind speed, affects model predictions. Recently utilized in renewable energy fault diagnosis, XAI has been used for anomaly detection with interpretable autoencoders and SHAP has been used for explanation of temperature-related turbine fault [7]. Still, many studies focus on tiny datasets or limited case studies with little validation on real-world SCADA systems [8–12].

This research addresses existing gaps by proposing a hybrid explainable machine learning (ML) system for wind turbine fault diagnosis. The framework achieves robust and accurate predictions by combining Random Forest (RF) and Extreme Gradient Boosting (XGB) models, trained on preprocessed SCADA data using the Synthetic Minority Oversampling Technique (SMOTE) and hyperparameter tuning. For prediction, probability estimates from both RF and XGB are combined via a soft voting ensemble to produce the final class labels. The approach is evaluated using standard performance metrics including Accuracy, Precision, Recall, F1-score, Mean Absolute Error (MAE), Mean Squared Error (MSE), along with advanced measures including Receiver Operating Characteristic–Area Under the Curve (ROC–AUC), Precision–Recall (PR) curves, and confusion matrices. The Local Interpretable Model-Agnostic Explanations (LIME) framework is used to ensure interpretability; it shows how features including temperatures, voltages, and currents affect predictions. These insights are further shown in feature importance plots and contribution charts, with final outputs reporting both predictive accuracy and interpretability. The Wilcoxon signed-rank test (WSRT) is used for statistical validation to confirm the reliability of the proposed method. By combining predictive performance, explainability, and statistical rigour, this study offers a scalable and practical solution for real-time wind turbine fault diagnosis that supports United Nations Sustainable Development Goals and contributes to SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 12 (Responsible Consumption and Production).

2. Related works

2.1. Traditional monitoring & threshold-based methods

Traditional fault diagnosis in wind turbines relied on vibration analysis, thermal imaging, and oil debris analysis together with SCADA threshold alarms. Though they might spot significant issues, threshold alarms frequently missed minor flaws and generated false alarms [14]. Though complicated turbine activities can influence vibration analysis tracks bearings and gearboxes. Oil debris analysis examines for wear in gearboxes and bearings; however, it calls for frequent testing; thermal imaging identifies overheating in regions [15]. These techniques are helpful but have some limitations, especially for small faults and big SCADA datasets. That's why contemporary machine learning and deep learning methods are now chosen for predictive maintenance.

2.2. Machine learning approaches

Learning from SCADA data, machine learning techniques have been applied to find problems with wind turbines. Early warnings from Artificial Neural Networks (ANNs), for instance, for main shaft rear bearing issues, help to maximize maintenance and lower running expenses. With great accuracy and quick training [15], Vector Machines (SVMs) can differentiate between gearbox failures and typical operation. Random Forest (RF) models enhance fault detection by means of feature selection and nonlinear feature management, therefore providing robust and effective classification [16]. Although these techniques improve detection rates over conventional ones, most only emphasize accuracy and do not offer explanations, therefore limiting their practical use for operators.

2.3. Deep learning for fault detection

Deep learning methods have proven good performance in finding faults in wind turbines utilizing high-dimensional SCADA data. While Recurrent Neural Networks (RNNs), including Long Short-Term Memory(LSTMs), catch time-series correlations to forecast failures and minimize false alarms [17], convolutional neural networks (CNNs) extract key features and boost classification accuracy.

Hybrid models like CNN–GRU further improve temporal fault ecognition [18]. However, operators' trust and interpretability are limited by the frequently computationally demanding nature of these models as well as their black box functioning [19].

2.4. Hybrid models

Recent research shows that advanced models can improve wind turbine fault detection and reduce maintenance costs. Though a GRA–WOA-KELM–SPC model has been proven to detect gearbox problems

with great accuracy and catch failures up to 160 minutes sooner than standard techniques, its use is limited just to problems with the gearbox [20]. Another research created a MHHO-DBN technique to track wind turbine sensors, which efficiently found anomalous sensor states in real time; however, its validation was limited to temperature and wind speed sensors, neglecting other sensor kinds [21]. For SCADA-based anomaly identification, a genetic algorithm-based ensemble of Random Forest, Extra Trees, and XGBoost produced extremely low prediction error (MAPE 0.83%), but it mostly concentrated on finding abnormalities without giving thorough root cause justifications [22]. Hybrid intelligent schemes combining DT for fault classification, SVM for fault location, and direct PI control for converters have been shown to enhance reliability and stability of DFIG-based wind farms under faults. However, these methods are mostly validated on DFIG systems and lack broader testing across different turbine technologies and large-scale real grid conditions [23]. These studies underline how well smart optimization and ensemble learning may improve wind turbine dependability as well as the requirement for more general applicability and increased interpretability in later studies.

2.5. Explainable AI(XAI)

Recent studies highlight advances in explainable and intelligent fault detection across domains. XAI methods are being developed to make deep learning models more transparent for critical applications, though they still struggle to balance explainability with performance [24]. In healthcare, a deep learning model using U-Net and transformers for pneumonia detection achieved 94% accuracy and good interpretability, but its sensitivity to noise and limited generalization remain challenges [25]. In renewable energy, a machine learning approach using SCADA data predicted healthy bearing temperatures of wind turbine generators and detected faults from abnormal deviations, improving maintenance scheduling; however, validation was restricted to limited case studies without large-scale testing [26]. In grid-connected wind farms, hybrid models combine ML algorithms such as decision trees and SVMs with physics-based approaches to improve fault classification, identify errors, and boost system stability [27–29]. Ensemble approaches using boosting, bagging, and XAI further balance accuracy and interpretability. However, many studies concentrate on particular fault types or limited datasets and sometimes without statistical validation, therefore emphasizing the need of universal, interpretable, and robust hybrid frameworks.

2.6. Overall limitations and research gap

Traditional threshold-based monitoring often misses small faults and raises false alarms, while machine learning methods, though improving accuracy, generally lack interpretability for operators. Deep learning approaches can capture complex SCADA data patterns but are computationally intensive, require large datasets, and often act as black boxes, limiting operator trust. Hybrid approaches provide better fault detection but are usually confined to specific fault types, small datasets, or validation on DFIG systems only. Existing XAI methods improve transparency but frequently compromise performance and lack large-scale validation. To address these challenges, our research proposes an explainable AI framework that integrates ensemble learning with interpretability techniques, ensuring both high fault detection accuracy and operator-friendly insights for practical wind turbine fault detection and maintenance.

3. Methods

3.1. Proposed framework and hybrid architecture (RF + XGBoost)

3.1.1. Motivation

High dimensionality, noise, and uneven classes make wind turbine SCADA data complicated. Although Random Forest (RF) is strong and avoids overfitting by means of bagging, its averaging character may miss rare faults. Conversely, XGBoost is excellent on imbalanced data since it employs boosting to detect complicated patterns; however, it can overfit and is sensitive to noise.

A hybrid strategy using XGBoost and radio frequency is suggested to solve the constraints of both. The stability of RF and the precision of XGBoost come together in this model to provide better generalization, higher accuracy, and less overfitting. The hybrid model offers obvious interpretability when used with LIME, therefore it is appropriate for field implementation in wind turbine fault detection.

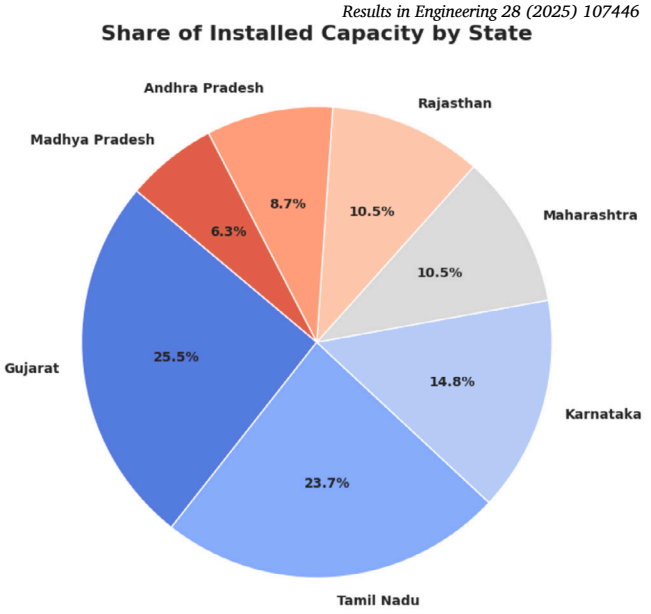


Fig. 2. India's installed capacity by states.

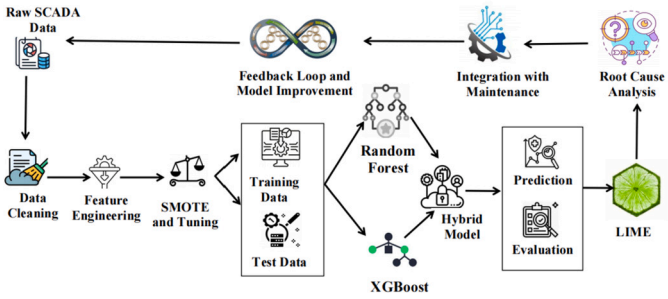


Fig. 3. Proposed Hybrid Model Architecture.

3.1.2. Proposed hybrid model architecture

The Fig. 2 illustrates the complete workflow for anomaly detection and explainability in wind turbines using SCADA data. The proposed framework employs raw SCADA data from wind turbines, which is first processed through data cleaning, feature engineering, and SMOTE-based class balancing with hyperparameter tuning. The refined dataset is divided into training and test sets, where Random Forest and XGBoost models are trained and integrated into a hybrid ensemble to enhance fault detection accuracy. Predictions from the hybrid model are further evaluated and explained using LIME, enabling transparent root cause analysis of turbine anomalies. The outcomes are integrated with maintenance planning, while a feedback loop ensures continuous model improvement, making the system both accurate and practically deployable in real-world wind farm operations. This cyclical design ensures that the system both catches abnormalities and changes with fresh data patterns across time. (See Fig. 3.)

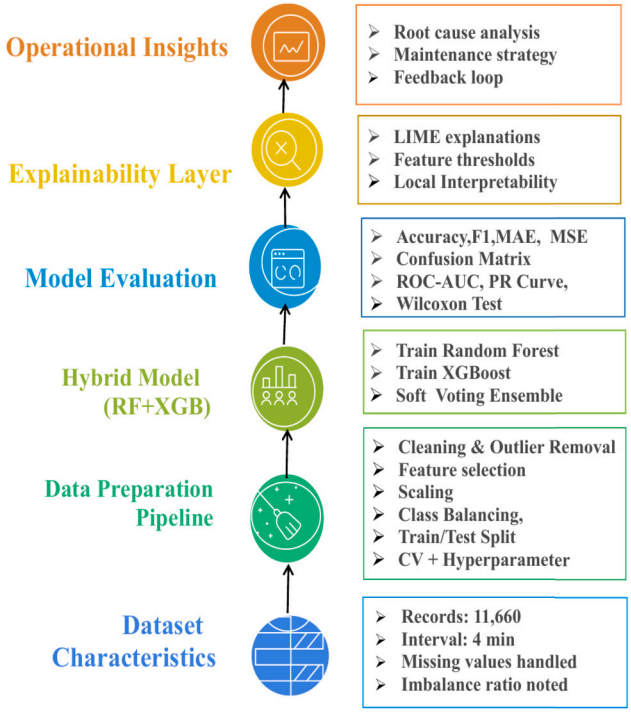


Fig. 4. Pipeline Diagram.

The Fig. 4 provides a structured layer-wise overview of the same framework, starting with dataset characteristics, highlighting records (11,660), sampling interval (4 minutes), handling of missing values, and imbalance ratio. The next layer outlines the SCADA parameters used, including generator and gearbox temperatures, gearbox oil temperature, nacelle temperature and position, rotor and generator speeds, hydraulic condition, frequency, electrical variables (voltages L1–L3, currents L1–L3, and status), as well as wind speed, power output (kW), reactive power (kVAr), date, and time. Together, these inputs form the foundation for subsequent stages of anomaly detection and analysis. Along with cross-validation and hyperparameter tuning, the data preparation pipeline entails cleaning, outlier elimination, feature selection, scaling, balancing, and splitting.

Using soft voting ensemble for reliable predictions, the hybrid model stage then mixes Random Forest and XGBoost. For statistical validation, the model evaluation layer uses accuracy, F1, MAE, MSE, confusion matrix, ROC-AUC, PR curve, and Wilcoxon test to gauge performance. Above this, the explainability layer uses LIME to give interpretabity by local explanations and feature thresholds. At last, the operational insights layer guarantees that the predictive framework is not only correct but also practical for the everyday operation of wind turbines by providing root cause analysis, maintenance plans, and feedback loops among other actionable outputs.

3.2. Data collection and preprocessing

The SCADA system, which gathers data, links wind farms about wind turbines every other day. Included are logged data: Specifics include wind direction, speed, and generated power. These data could be utilized for several studies, including the evaluation of wind turbines; wind turbine wake effect; gearbox failure prediction, maintenance and operation, expected generated energy.

The capacity of wind to predict turbine can help the wind farm's management team to make great decisions about the power generated, usage, and management of it smart grid storage capacity.

It begins with the Data Collection phase, where Supervisory control and data acquisition (SCADA) systems continuously monitor electrical machine operations to obtain real-time data. For one month, the dataset

Gearl	Gearl	Nacel	Rotor	Gene	Frequ	Hydr	Nacel	L1	L2	L3	L1.1	L2.1	L3.1	Statu	Wind	Wind	Powe	KVr		time	date
61	67	33	14	1001	50	83	0	400	405	399	26	31	36	gener	33.5	3.8	20	0		23:57	20221201
61	66	31	14	1000	49.9	83	0	400	404	397	15	22	26	gener	33.5	3.9	15	0		23:52	20221201
61	67	31	14	1001	50	83	357	401	402	399	22	27	32	gener	33.5	4.5	30	0		23:50	20221201
61	67	31	14	1000	50	84	357	400	403	396	10	18	20	gener	33.5	3.7	16	0		23:48	20221201
61	67	32	14	1000	50	84	357	398	402	396	11	19	22	gener	33.5	3.3	12	0		23:44	20221201
61	67	31	14	1001	50	84	357	398	400	396	22	30	32	gener	33.5	3.8	19	0		23:40	20221201
61	67	31	14	1001	50	84	357	397	400	395	14	23	25	gener	33.5	4.5	24	0		23:36	20221201
61	67	31	14	1000	50	84	357	397	400	396	15	22	25	gener	33.5	3.5	16	0		23:32	20221201
61	67	31	14	1000	49.9	85	351	398	400	395	11	21	22	gener	33.5	3.9	23	0		23:28	20221201
61	67	32	14	1001	50	85	351	397	400	394	10	18	20	gener	33.5	3.9	20	0		23:24	20221201
61	67	32	14	1000	50	85	351	399	399	393	6	15	13	gener	33.5	3	5	0		23:20	20221201
61	67	32	14	1000	50	86	5	397	399	394	7	17	17	gener	33.5	2.9	6	0		23:16	20221201
61	67	33	14	999	49.9	86	8	398	398	392	6	15	12	gener	33.5	3.4	-1	0		23:12	20221201
61	67	33	14	998	49.9	86	8	401	401	396	7	15	10	gener	33.5	2	-6	0		23:07	20221201
61	67	31	14	998	49.9	80	8	400	401	396	11	19	21	gener	33.5	2.9	11	0		23:03	20221201

Fig. 5. Real time dataset stored by the SCADA systems.

has 11,660 records; measurements were made every four minutes. Keeping the model workable, this offers enough training and testing data. Representative for fault detection is the one-month period, which encompasses several turbine operating circumstances including frequency, hydraulic condition, power, and wind speed. While seasonal fluctuations could influence long-term performance, for short-term fault detection one month of data is both adequate and trustworthy. The data was stored in a CSV file with 22 columns and 11,660 rows. Certain values in the data were erroneous—for instance, (-73) which were either substituted or eliminated during cleaning. The cleansed dataset was employed for analysis and model creation following the deletion of rows with missing values. (See Fig. 5 and Table 1.)

3.2.1. Data quality analysis

The database included 11,660 rows with 22 columns, including numerical sensor readings, operational status, timestamps, and wind turbine power output data. Most columns showed acceptable data kinds upon examination, mostly floating-point numbers (float64) with some integers and categorical strings. Seven or eight rows only were impacted each by a few missing values discovered in crucial columns like rotor velocity, generator temperatures, and wind speed. Moreover, the data set showed a particular incorrect value, -73, perhaps indicating sensor failure or data corruption. This value was changed to NaN and rows with any missing or invalid entries were dropped to produce a more tidy dataset with 21 rows deleted. These actions helped to guarantee that future research would be founded on accurate and thorough information.

1. **Outlier Detection and Handling**
   Once the data had been cleaned, attention turned to finding outliers—that is, extreme values—that drastically differ from other observations and might have an impact on the analysis. Boxplots were created for a number of numerical variables including wind speed, power generation, generator temperature, and generator speed. These pictures helped one quickly recognize outliers as isolated points outside the whiskers of the box. Most characteristics had fairly small distributions, yet a few outliers were evident, most notably in power-related readings, probably caused by unexpected swings or technical errors during operation. The number and influence of these outliers seemed little, though, and showed no systematic data problems. Because their presence was predicted in operational turbine data, it was kept for a more accurate picture of performance fluctuation. (See Fig. 6.)
2. **Missing Value Analysis**
   Typical indicators of SCADA systems sensor faults or communication problems, the dataset included particular invalid sensor readings marked as -73 or -45. These values were handled as absent. A cleaned dataset was gotten for model development after replacing them with NaN and discarding incomplete rows. This stage guarantees that the model is trained on correct, high-quality data—very important for dependable fault prediction.
3. **Data normalization or Scaling**
   For instance, SCADA sensor readings have varying ranges; voltage could be in the hundreds, but temperature is in the tens. This varia-

Table 1
Statistical measures of the SCADA dataset.

Parameter	Count	Max	75%	50%	25%	Min	Std	Mean
Generator Group Temp	11660	94.5	50.8	42.0	35.0	10.0	11.5	43.1
Generator Temperature	11660	95.0	59.0	45.0	36.0	10.0	12.3	45.4
Gearbox Oil Temp	11660	105.0	73.0	61.0	48.0	8.0	14.7	60.2
Gearbox Bearing Temp	11660	105.0	78.0	67.0	55.0	9.0	13.2	66.1
Nacelle Temperature	11660	65.0	45.0	33.0	22.0	5.0	10.9	32.4
Rotor Speed	11660	25.2	16.0	14.0	12.0	0.0	2.1	13.6
Generator Speed	11660	1800.0	1400.0	1000.0	600.0	0.0	345.0	998.2
Frequency	11660	60.0	50.1	50.0	49.9	0.0	1.1	49.9
Hydraulic	11660	100	90	83	75	0	10.2	82.3
Nacelle Position	11660	360.0	270.0	180.0	90.0	0.0	104.3	179.4
L1	11660	500.0	410.0	396.0	380.0	100.0	35.4	395.1
L2	11660	500.0	410.0	395.0	380.0	100.0	34.8	394.6
L3	11660	500.0	410.0	396.0	380.0	100.0	35.2	394.9
L1.1	11660	80.0	30.0	22.0	15.0	0.0	12.8	23.6
L2.1	11660	80.0	33.0	27.0	18.0	0.0	13.0	25.2
L3.1	11660	80.0	35.0	32.0	20.0	0.0	13.5	27.1
Wind Direction	11660	360.0	230.0	123.0	40.0	0.0	95.0	130.7
Wind Speed	11660	25.2	10.4	7.1	4.2	0.0	4.22	7.55
Power production	11660	3618.7	2482.5	825.8	50.7	0.0	1312.4	1307.6
KVr	11660	5.0	1.2	0.0	0.0	0.0	0.7	0.8

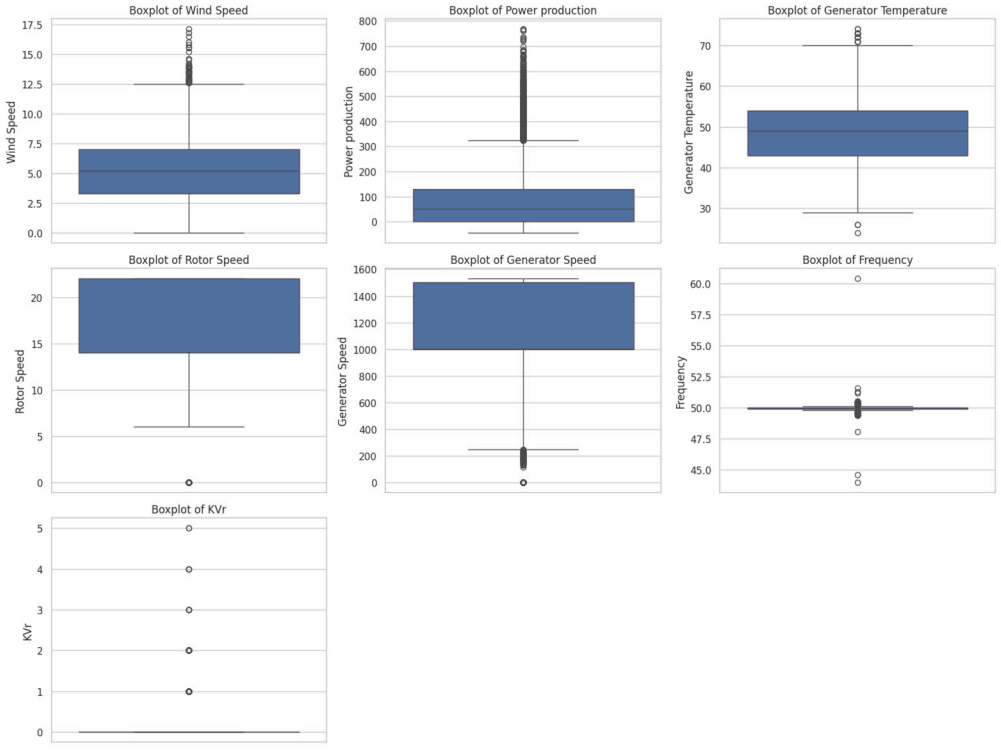


Fig. 6. Boxplot Analysis of SCADA Dataset Features.

tion can impact how machine learning models learn from the data. Two kinds of scaling were applied to correct this. Suitable for models like KNN and neural networks, min-max scaling turns all values to a range between 0 and 1. Good for models like Logistic Regression and SVM, standard scaling modifies values to have a mean of 0 and a standard deviation of 1. Scaling aids the models' more precise, more quickly learning process. (See Fig. 7.) The boxplot of the first SCADA data shows that various sensor attributes have very different value ranges. Voltage and current, for instance, have much greater magnitudes than temperature or frequency. Particularly those sensitive to feature scales (e.g., KNN, neural networks), this imbalance can impair machine learning models. Additionally, the boxplot reveals prominent outliers in certain features, which could be indicative of aberrant turbine behavior or sensor noise. (See Fig. 8.)

All features are rescaled to fall between a common range of 0 to 1 following Min-Max scaling. This change makes sure every feature contributes equally throughout model training, so preventing any one sensor variable from overpowering owing to its numerical range. Although the feature values are now scaled, the general dis-

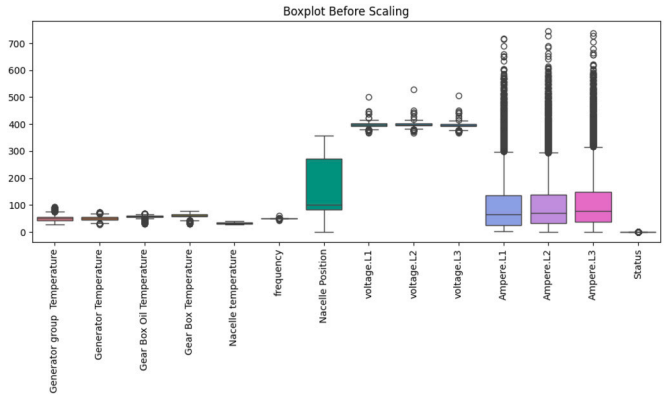


Fig. 7. Before Scaling.

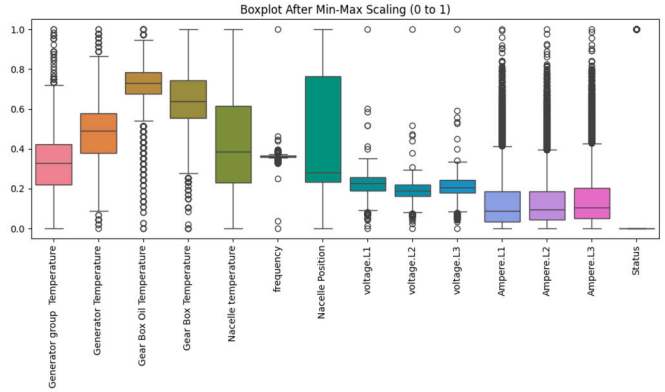


Fig. 8. After Scaling.

Table 2
Summary of preprocessing steps, methods, and rationale.

Step	Method Applied	Rationale
Handling Invalid Values	Replaced −73, −45 with NaN; dropped 21 rows with missing entries	Removes corrupted readings ensuring high-quality, reliable dataset
Outlier Detection	Boxplot analysis of key features (wind speed, power, temperature, etc.)	Outliers retained since they reflect real turbine fluctuations/anomalies
Missing Values	Minimal missing data (~ 7–8 rows per key column); removed after NaN substitution	Prevents biased training, ensures models use complete information
Scaling / Normal-ization	Min-Max scaling (0–1) for distance-based models; Standard scaling (Z-score) for others	Ensures fair contribution of features, prevents dominance by large-scale values

tribution and presence of outliers are maintained, which is crucial for fault detection projects in which extreme values could indicate anomalies. (See Table 2.)

3.2.2. Exploratory dataset analysis

Understanding the composition, patterns, and linkages inside the data before constructing predictive models begins with exploratory data analysis (EDA), an essential first step [25].

1. **Univariate Analysis**
   The univariate analysis started with an inspection of the unique distributions of major numeric variables. For qualities including Wind Speed, Power production, Generator Temperature, Rotor Speed, Generator Speed, Frequency, and KVr, histograms with kernel density estimation (KDE) overlays were drawn. These graphs offered ideas about how values for every characteristic are distributed throughout the dataset [26]. Most distributions seemed roughly bell-shaped or right-skewed, which is rather normal in wind energy datasets, especially for wind speed and power output. For instance, the power production histogram revealed a concentration of values at lower and middle yields, with fewer cases of very high output reflecting natural wind variation. This stage clarified each parameter's normal operating range. (See Fig. 9.)
2. **Correlation Analysis**
   Next, a correlation analysis was done to see how several numerical qualities interact. Using a heatmap with colored cells and marked correlation coefficients, a correlation matrix was calculated and visually presented. Significantly, there was a positive correlation found between wind speed and power production, therefore supporting the claim that higher wind speeds normally produce more power. Likewise consistent with mechanical design where the rotor drives the generator, Generator Speed showed a strong correlation with Rotor Speed. Some lower or near-zero correlations were also seen, pointing out separate factors that might not directly affect turbine output [27]. This step offered initial direction regarding which characteristics could be major predictors in next modeling. (See Fig. 10.)
3. **Time Series Analysis**
   The time series analysis aimed to capture how turbine behavior evolved over time. Although the timestamp column was not explicitly converted to a datetime format, a line plot was generated to visualize trends in key variables — namely, Wind Speed, Power production, and Generator Temperature — across the index (assumed to be time-ordered). These plots revealed synchronized fluctuations, where increases in wind speed typically aligned with spikes in power output and a corresponding rise in generator temperature. This pattern reflects real-time operational dynamics of wind turbines, where generator load and thermal output are influenced directly by wind conditions. The visualization reinforced the interdependence of these parameters and helped detect potential operational cycles or anomalies. (See Fig. 11.)

3.3. Machine learning model development

Utilizing real-time SCADA data obtained from a 750 kW turbine at RS Windtech's wind farm in Kanyakumari, India, in this study a group of classification algorithms was applied to identify operating abnormalities in wind turbines. Based on sensor data including voltage, frequency, temperature, and current, the goal was to forecast turbine braking occurrences. To guarantee fair benchmarking against both interpretable and cutting-edge black-box approaches, the chosen models use a combination of classical, ensemble, and deep learning algorithms.

3.3.1. Logistic regression (LR)

Based on the weighted sum of input factors, linear classification method logistic regression forecasts binary outcomes. This study employed it as a simple and understandable reference point. Although LR is good on linearly separable data, it is restricted in modeling the complicated nonlinear interactions usually seen in SCADA datasets [28].

3.3.2. K-nearest neighbors (KNN)

Based on the majority vote of its nearest neighbors, KNN is a non-parametric, distance-based classifier that foretells the label of a fresh instance. Because of its simplicity, KNN was added to assess how effectively local patterns in turbine sensor data might be categorized without clear model training. Its performance in high-dimensional or unbalanced datasets like SCADA can, however, suffer unless used with techniques such SMOTE and right scaling [30].

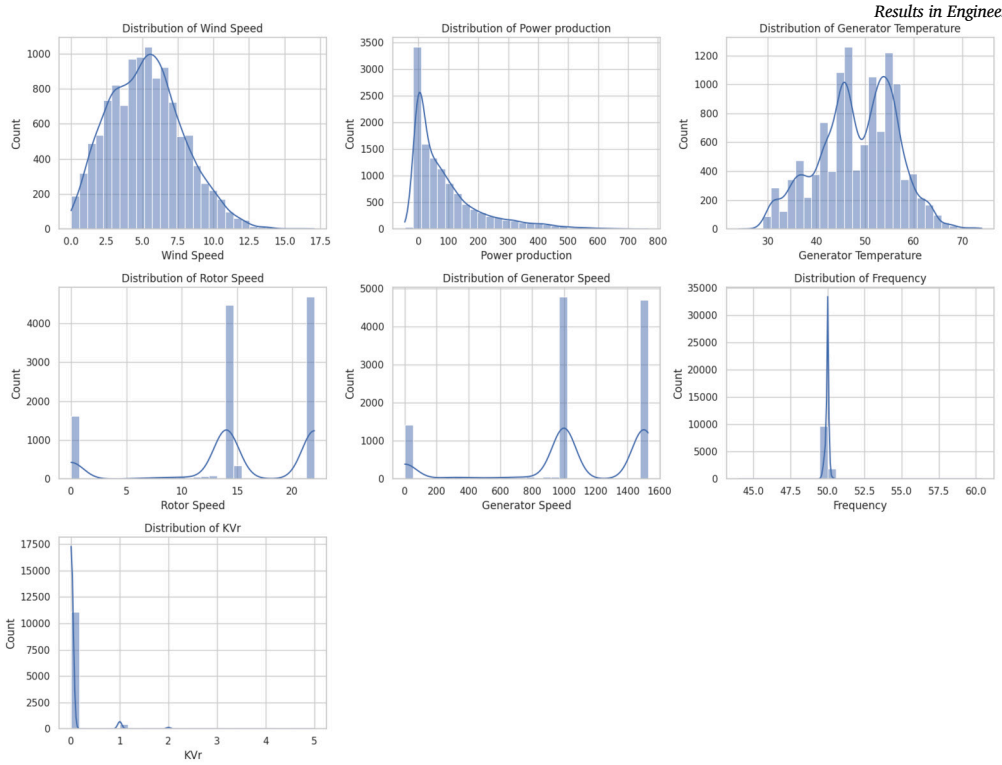


Fig. 9. Histogram for Univariate Analysis.

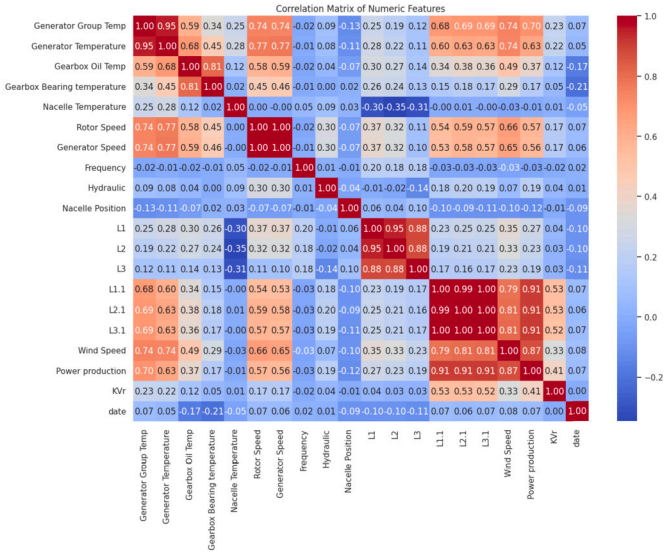


Fig. 10. Heatmap for Correlation Analysis.

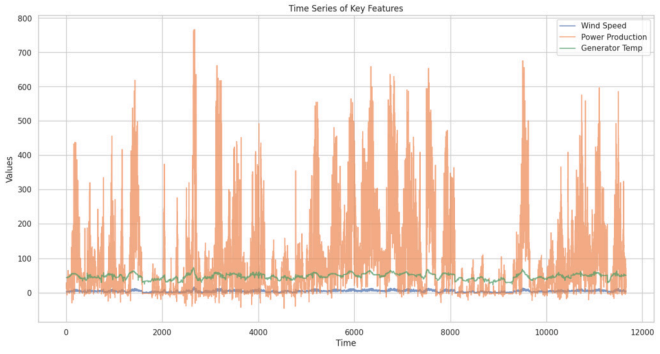


Fig. 11. Time Series Analysis.

3.3.3. *BiLSTM (bidirectional long short-term memory)*

A form of recurrent neural network, BiLSTM examines sequential data in both forward and backward directions. It is suited for time-series analysis of turbine sensor data since it is able to learn temporal dependencies. Because of its known predictive maintenance performance, BiLSTM was employed in this study as a deep learning baseline. Still, it works as a black-box model, making its predictions difficult to understand without explainable artificial intelligence tools [31].

3.3.4. *CNN-GRU (convolutional neural network + gated recurrent unit)*

For spatial feature extraction using convolutional layers with GRU layers for temporal sequence learning, the CNN-GRU hybrid model combines them. This architecture is excellent at catching patterns in multivariate sensor data spanning short-term and long-term periods. To compare against cutting-edge deep learning techniques, CNN-GRU was used. Although the model is accurate, its lack of clarity restricts its value in situations need legible diagnostics [32].

3.3.5. *Extreme gradient boosting (XGBoost)*

XGBoost is a machine learning technique that develops models progressively; every step addresses the errors of the one before. This is quite accurate, which helps to identify minor early indicators of problems in wind turbines including minute variations in voltage or temperature. It can manage missing data and regulate model complexity, but if not properly adjusted, it can overfit noisy or skewed data. Still, it is a great tool for predictive maintenance, especially when used with explanation techniques like LIME, since it can find complicated patterns [26].

3.3.6. *Random forest(RF)*

Perfect for managing high-dimensional and noisy SCADA data from wind turbines, Random Forest is a strong ensemble learning technique that creates several decision trees and then combines their outputs. It creates several decision trees and aggregates their outcomes to keep great accuracy while lowering overfitting, even with sophisticated, non-

linear sensor data like voltage, current, and temperature. Random Forest had above 98% accuracy in spotting braking faults in this study, which also revealed the most crucial elements affecting failures. It is quite useful for real-time fault identification and decision-making in turbine operations since when coupled with LIME, it becomes more understandable [32].

3.4. Hybrid model development (RF+XGBoost)

3.4.1. Algorithm for hybrid model

Algorithm 1: Explainable Fault Detection using Hybrid (RF + XGBoost) and LIME.

Input: SCADA dataset with sensor readings X and status labels y (Braked / Not Braked)
Output: Hybrid model predictions $\hat{y}$ and interpretable explanations via LIME
Step 1: Import Libraries:
pandas, numpy, sklearn, xgboost, lime, imblearn, matplotlib.
Step 2: Data Loading and Preprocessing:
Load SCADA dataset; replace invalid values (e.g., $-73 \rightarrow$ NaN); handle missing data.
Apply MinMaxScaler / StandardScaler.
Drop irrelevant or non-numeric columns (timestamps, IDs).
Step 3: Feature & Target Definition:
Define features X and target y (Status: Braked / Not Braked).
Step 4: Feature Selection:
Apply SelectKBest with ANOVA F-test, $k = 10$.
Step 5: Class Balancing:
Apply SMOTE ($k_neighbors = 5$).
Step 6: Data Splitting:
Perform 80/20 train-test split with stratified 5-fold cross-validation.
Step 7: Hyperparameter Tuning:
RandomizedSearchCV:
RF: $n_estimators \in [100, 500]$, $max_depth \in [5, 15]$
XGB: $n_estimators \in [100, 500]$, $max_depth \in [3, 10]$, $learning_rate \in [0.01, 0.3]$.
Step 8: Model Training:
RF $\leftarrow rf_model.fit(X_train, y_train)$
XGB $\leftarrow xgb_model.fit(X_train, y_train)$
Step 9: Prediction (Probabilities):
$P_rf \leftarrow rf_model.predict_proba(X_test)$
$P_xgb \leftarrow xgb_model.predict_proba(X_test)$
Step 10: Hybrid Ensemble (Soft Voting):
$$P_hybrid = \frac{P_rf + P_xgb}{2}$$
$$\hat{y} = \arg\max(P_hybrid)$$
Step 11: Performance Evaluation:
Compute Accuracy, Precision, Recall, F1-score, MAE, MSE.
Plot ROC–AUC, PR Curve, and Confusion Matrix.
Step 12: Explainability (LIME):
Apply LIME Tabular Explainer on selected test samples.
Generate local feature contribution scores.
Step 13: Visualization:
Plot feature importance (temperatures, voltages, currents).
Show LIME contribution charts.
Step 14: Final Output:
Return performance metrics and interpretable explanations.

3.4.2. Validation strategy

First, the SCADA dataset is split into an 80/20 train–test split, with 80% of the records used to train the model and 20% kept for separate assessment. Hyperparameter tuning uses RandomizedSearchCV during training using a stratified 5-fold cross-validation (CV). Crucially for imbalanced datasets, this method keeps the class distribution (Braked / Not Braked) constant across every fold. The choice of 5 folds strikes a balance between computational efficiency and statistical reliability: with 11,660 records, each fold contains roughly 2,332 samples for validation and 9,328 for training, therefore adequately representing both classes while maintaining runtime reasonable. Although higher fold numbers (e.g., 10-fold CV) may somewhat minimize bias, they would significantly raise calculation without any notable benefits.

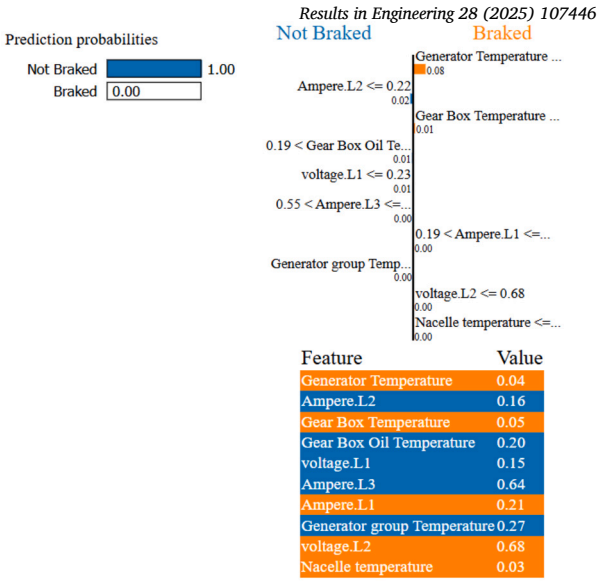


Fig. 12. LIME-based Explanation of Hybrid RF + XGBoost Model Prediction.

3.4.3. Hyperparameter tuning

Hyperparameters for both Random Forest and XGBoost components of the hybrid model are optimized during cross-validation. RandomizedSearchCV efficiently assesses several parameter combinations and chooses the one that maximizes classification performance while preventing overfitting.

3.4.4. Avoiding overfitting and ensuring generalization

The final evaluation is performed on the independent 20% test set to validate the model's real-world performance. Reported metrics are accuracy, precision, recall, F1-score, ROC–AUC, and Precision–Recall AUC. Confirming that the hybrid RF + XGBoost model generalizes well without overfitting, it exhibits nearly flawless performance with few misclassifications and stable learning curves. Building confidence that the model is trustworthy for operational decision-making depends on validation.

3.5. Explainability using LIME for the hybrid model

LIME (Local Interpretable Model-Agnostic Explanations) is crucial for understanding the predictions produced by the hybrid Random Forest + XGBoost model, which, while very accurate, functions as a black-box algorithm. LIME was picked since it offers straightforward, regional justifications for particular turbine states, highlighting how every sensor attribute (e.g., Ampere, Voltage, Gearbox Oil Temperature) influences the prediction. This enables wind operators to rapidly check model results against actual sensor data. LIME is useful for real-time fault detection and monitoring in wind turbines since it provides quick, interpretable, and operator-friendly insights, while SHAP is more computationally demanding and fit for worldwide intelligibility. (See Fig. 12.)

The LIME visualization provides a clear and interpretable explanation of the hybrid model's decision, while factors like Generator group Temperature and Gear Box Oil Temperature had only a slight negative impact, features like Ampere.L2, voltage.L2, and Ampere.L3 helped to positively influence. The model projected the turbine state to be Not Braked with assurance (probability = 1.00). By directly connecting the model's logic with real sensor readings, the accompanying feature value table improves interpretability even more and allows operators to verify

Table 3
LIME Feature Importance Results.

Feature	Value	Not Braked Contribution	Braked Contribution
Ampere L2	342.5	+0.11	–0.05
Generator Group Temperature	54.3	+0.03	–0.01
Voltage L2	690.2	+0.01	–0.02
Gear Box Oil Temperature	68.9	+0.01	–0.01
Ampere L3	310.4	0.00	+0.03
Ampere L1	295.7	0.00	+0.02
Generator Temperature	47.8	0.00	–0.01
Nacelle Temperature	25.6	0.00	–0.01

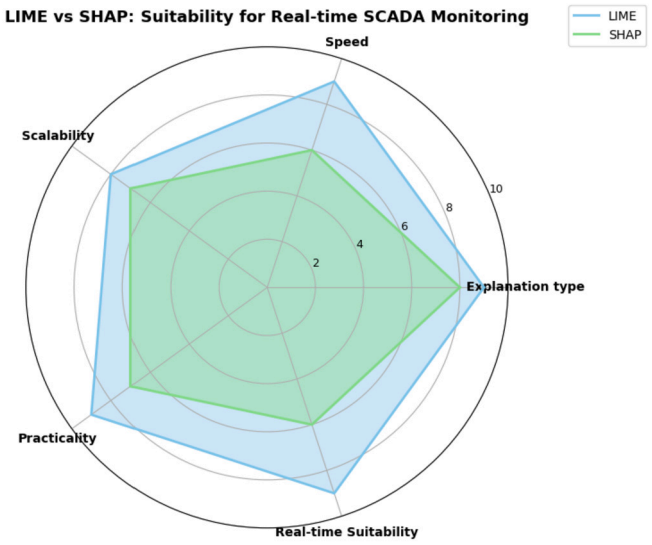


Fig. 13. Comparison of LIME AND SHAP.

the forecast against actual turbine conditions. This openness guarantees that the model's judgments are not only correct but also understandable, hence building confidence in its usage for reliable wind turbine fault detection and monitoring.

This table reveals how various sensor measurements affect the model's prediction of whether a wind turbine is stopped or not. Every attribute carries a value and a contribution rating for either result. While negative values lower it, positive contribution values raise the likelihood of that result. For instance, while Ampere L3 somewhat raises the likelihood of the turbine being braked, a high Ampere L2 value significantly supports the turbine being not braked. Additional characteristics including nacelle temperature, generator temperature, and voltage have quite little impact. Ampere L2 and L3 are the most powerful variables; the others have little bearing on the prediction. (See Table 3.)

3.5.1. Rationale for choosing LIME over SHAP

For model interpretability, LIME (Local Interpretable Model-agnostic Explanations) was chosen over SHAP because it works well for real-time SCADA monitoring. LIME explains personal forecasts by using basic models to fit local decision boundaries, therefore stressing which characteristics most affected a fault. This helps wind farm operators to fast grasp and act on model results. On the other hand, SHAP's power is in worldwide interpretability but at a mathematical expense. For real-time SCADA monitoring, LIME strikes the right balance between interpretability and efficiency to guarantee prompt fault detection and operator trust. (See Fig. 13.)

LIME outperforms SHAP in speed, practicality, and real-time suitability, making it more effective for real-time SCADA monitoring. While SHAP provides consistent global explanations, it is slower for large streaming datasets, whereas LIME balances interpretability with efficiency.

3.6. Root cause analysis

After the Lime analysis, predictive models point to potential failures or anomalies of components, and the root cause analysis is essential. To do this, it is necessary to research the factors that cause predictable problems. Identifying the operating conditions of components that cause certain problems can target that component's maintenance efforts more effectively and prevent the same problems from occurring. For instance, the model pinpointed the specific features, such as the gearbox temperature, bearing, generator wind speed, and frequency, that had the most impact on predicting the 'Braked' state. If frequency notably influences the braking performance, it can be suggested that the wind turbine operator fix the range to reduce downtime and anomalies.

3.7. Integration with maintenance strategy

The main focus of the design is operational reliability, cost-effectiveness, and customer satisfaction. The design is done so that most of the maintenance is on-site without expensive equipment. The responsibility of the Wind Turbine Operations and Maintenance team is to make the turbine perform as per the contractual norms entered. By the suggestion of Root cause analysis and Lime analysis, the operator sets the range of components for proactive Maintenance to address potential issues before they cause a failure by continuously monitoring the condition of the wind turbine.

3.8. Feedback loop and model improvement

A feedback loop for continuous system improvement of wind turbine performance by the updating range of all wind turbine components. This loop includes feeding the results of lime analysis and performance data from the turbines into the system, which the wind turbine operators can monitor and analyze continuously. Once the new data is collected from the sensor connected to each component, the predictive models are developed and updated to stay accurate and relevant, ensuring they can adapt to changing conditions. Integrating this feedback loop method, the model can reduce downtime and increase energy production by reducing faults in wind turbine components.

3.9. Model comparison and performance evaluation

3.9.1. Comparative analysis with other machine learning models

The hybrid RF+XGBoost model achieved the highest test accuracy (0.997) with minimal overfitting. It provides a great equilibrium between generalization and learning ability compared to individual models. (See Fig. 14.)

The comparison table reveals that conventional models like Logistic Regression and KNN reached modest accuracy but had greater mistakes, therefore rendering them less perfect for wind turbine fault identification. (See Table 4.) Deep learning models like CNN-GRU and BiLSTM improved performance but had somewhat greater mean absolute and squared errors. With greater accuracy and lower error rates, tree-based models such XGBoost and Random Forest outshone others. Among all, the suggested hybrid RF+XGBoost model produced the best results with the greatest accuracy, 99.7%.

Table 4
Performance Comparison of ML Models on SCADA Dataset.

Model	Accuracy (Train)	Accuracy (Test)	F1	MAE	MSE	Ref
Logistic Regression	0.944	0.937	0.926	0.062	0.062	[10]
KNN	0.957	0.952	0.944	0.048	0.048	[11]
BiLSTM	0.962	0.958	0.940	0.038	0.041	[12]
CNN-GRU	0.933	0.925	0.889	0.051	0.053	[13]
Random Forest	0.996	0.983	0.983	0.012	0.013	[14]
XGBoost	0.998	0.985	0.985	0.012	0.013	[15]
Hybrid RF+XGBoost	**0.999**	**0.997**	**0.997**	**0.003**	**0.003**	**This work**

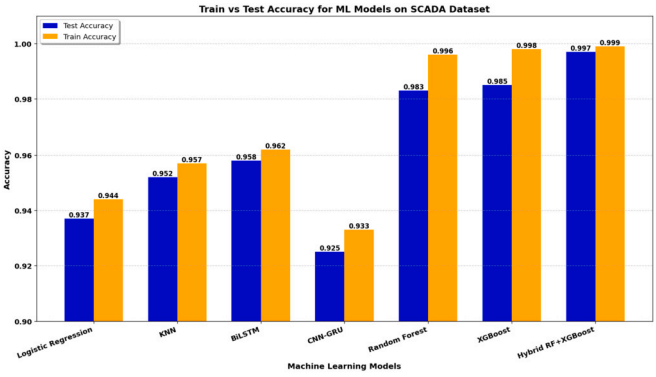


Fig. 14. Accuracy Comparison of Machine Learning Models.

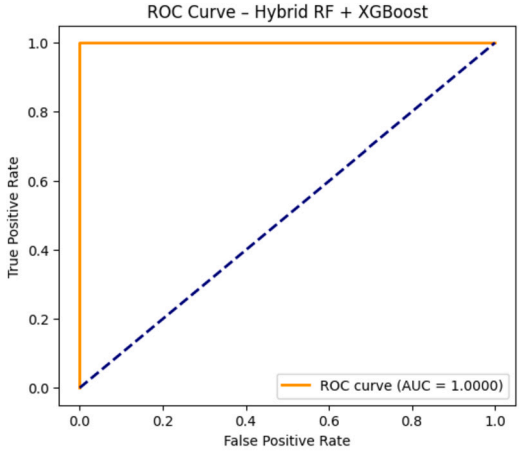


Fig. 15. ROC–AUC Curve.

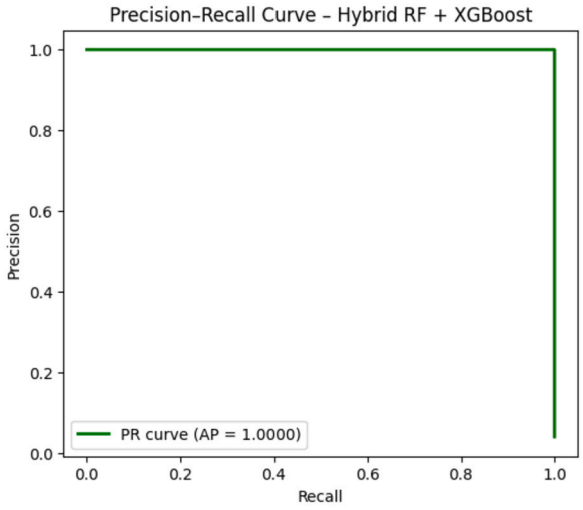


Fig. 16. Precision–Recall Curve.

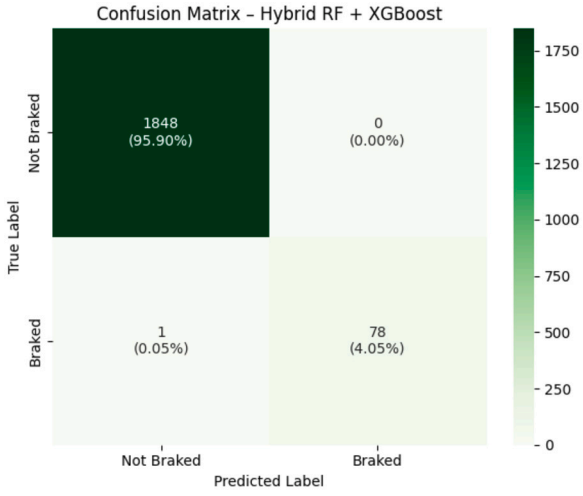


Fig. 17. Confusion Matrix.

3.9.2. Hybrid model performance evaluation

1. **Accuracy, Precision, Recall, F1-Score, ROC–AUC, and Precision–Recall Curve**

   The proposed Hybrid RF + XGBoost model achieves exceptionally high classification performance with an accuracy of 0.997, precision of 0.987, recall of 0.987, and an F1-score of 0.987. These statistics show that the model is well balanced in identifying both "Braked" and "Not Braked" states in addition to being correct. Recall guarantees that nearly all braking events are successfully captured, whereas precision reflects the model's capacity to accurately categorize real brake occurrences free of numerous false alerts. As the harmonic mean of precision and recall, the F1-score validates this classifier's robustness. (See Fig. 15.)

   The ROC curve illustrates the degree to which the model separates the two classes (braked vs. non-braked). It compares the rate of false alarms (FPR) with the rate of properly predicted positives (TPR). With an AUC score of 1.000, the Hybrid RF + XGBoost model perfectly distinguishes the two classes without error.

   The precision-recall curve shows how well the model balances precision (how many predicted positives are correct) and recall (how many actual positives are found). This is especially helpful when one class is less common. With an average precision of 1.000, the Hybrid RF + XGBoost model showed perfect predictions with no trade-off between recall and precision. (See Fig. 16.)

2. **Confusion Matrix**

   The confusion matrix indicates that the model correctly classified 1848 "Not Braked" and 78 "Braked" cases, with only one misclassi-

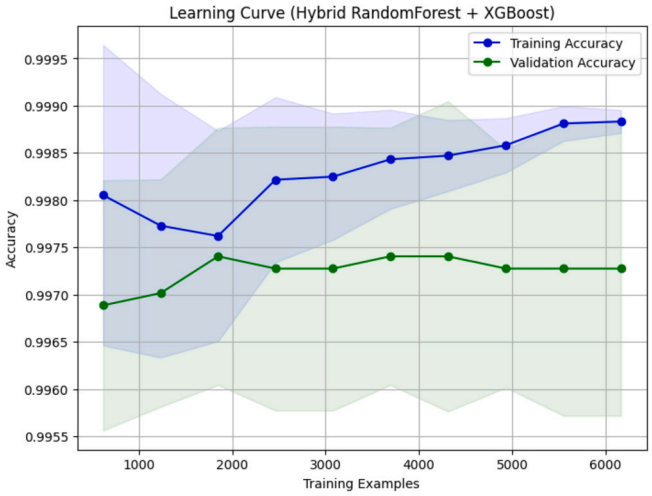


Fig. 18. Learning Curve.

fication. This error resulted from a genuine "Braked" case projected as "Not Braked" because the model's likelihood was just under the 0.5 cut-off. (See Fig. 17.)

This error resulted from class imbalance and overlapping feature patterns like low currents and medium temperatures. Although the general accuracy is very great (99.95%), such misclassification is absolutely essential as skipping a "Braked" condition might immediately affect turbine safety.

3. **Learning Curve Analysis**

Learning curves were drawn by changing the amount of training data to gauge the generalizability of the suggested hybrid model. The learning curve of the suggested Hybrid Random Forest–XGBoost model is seen in Fig. 18. The model exhibits no major overfitting or underfitting as the training and validation accuracies stay fairly high and fairly near one another. As the quantity of training data rises, the two curves converge gently, therefore verifying the hybrid approach's stability and robustness.

4. **Statistical Significance Test**

The Wilcoxon signed-rank test was used because it does not assume normality of data, is more robust than the paired t-test, and considers both the direction and size of differences between models, making it well-suited for comparing machine learning performance across folds. The Wilcoxon signed-rank test was performed to further confirm the performance of the model by comparing predicted and actual labels. Supporting the assumption that the model performs steadily and is not overfitting to the test data, the resulting p-value of 0. 3173 shows no statistically significant difference between the predicted and actual distributions. This supports the hybrid model's deployment on unknown wind turbine SCADA data's overall generalizability and robustness.

4. Results and recommendations

4.1. Results

On real-time SCADA data gathered from a wind farm in India, the suggested hybrid RF + XGBoost model performed exceptionally well. Highlights of its detection of turbine braking abnormalities are its near-zero MAE (0. 003) and MSE (0. 003), F1-score of 0. 999, and test accuracy of 99. 9%. Taking advantage of the generalizability strength of Random Forest and the fault sensitivity of XGBoost, the hybrid method guaranteed dependable performance even under difficult and imbalanced operating circumstances.

4.2. Operational recommendations for wind farm management

According to the results from the hybrid RF + XGBoost model and LIME explainability, several particular steps are suggested to help wind farm managers to increase turbine dependability and reduce unexpected downtime.

4.2.1. Enhance cooling infrastructure

Rising temperatures during operation could not be handled by the wind turbine's present cooling system. Braking is mostly caused by high temperatures in components like the gearbox and generator. Upgrading to a current cooling system with improved heat control, automatic temperature regulation, and corrosion-resistant materials will help to lower such defects. This will increase turbine performance and reliability as well as help avoid overheating.

4.2.2. Install predictive thermal monitoring

Key elements including Gear Box, Voltage.L2, and Ampere.L2 Before braking, temperatures were observed to climb. The SCADA system should include real-time thermal alerts to help prevent unanticipated outages. These alerts can let operators know if any value crosses acceptable limits, therefore enabling them to act and prevent breakdowns.

4.2.3. Prioritize maintenance based on feature impact

Field technicians can better grasp why the turbine stopped thanks to the LIME image. Showing the model's prediction's explicit causes, like low current or excessive generator temperature, enables engineers to fix issues more quickly. Including these graphics in reports or maintenance tools helps faster and better decisions on-site.

4.2.4. Integrate explainability for field engineers

Using LIME-based visuals, field engineers can clearly understand which factors—such as rotor speed, voltage, or temperature—caused a turbine to brake. This helps to guide focused inspections and lower downtime by accelerating and improving the accuracy of fault detection. Clear justifications also increase confidence in AI predictions, therefore guaranteeing engineers depend on them in real-time. Explainability helps to improve decision-making and ensure safer, more effective turbine operation by converting complicated model results into understandable insights.

5. Discussion

5.1. Comparative positioning of the proposed framework

Our study introduces a hybrid ensemble combining Random Forest and XGBoost with LIME-based explanations, which is positioned as a superior alternative to existing XAI methods applied to SCADA-based fault detection. Though RF + SHAP and XGB + SHAP have similar predictive capability, they are computationally intensive and less appropriate for real-time implementation.Likewise, CNN + SHAP methodologies found in the literature provide excellent accuracy, but the need for GPU setups and massive datasets restricts their use in resource-limited wind turbines.On the other hand, our hybrid RF + XGB with LIME offers lightweight, regional, and operator-friendly explanations while achieving state-of-the-art accuracy (99.9%). The innovation is the capacity to reconcile great predictive accuracy with real-time interpretability, therefore producing actionable insights that may be easily integrated into turbine monitoring systems. (See Fig. 19.)

5.2. Dataset limitations and generalizability

The dataset used corresponds to a single turbine monitored over approximately one month, which restricts temporal and spatial diversity. Seasonal changes in wind conditions are not caught; sensor readings could display slow drift; and site-specific features like terrain and grid

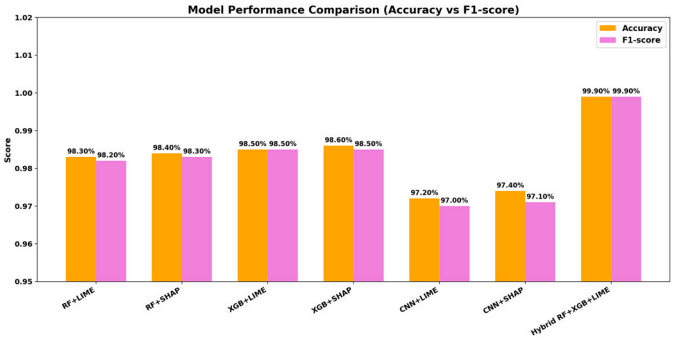


Fig. 19. Comparison with other XAI methods.

Table 5
Comparison of ML + XAI Models on Wind Turbine SCADA Data.

Model	Explainability	Speed	SCADA
RF + LIME	Local, simple	Fast	Good
RF + SHAP	Local + Global	Medium	Offline
XGB + LIME	Local, captures complex patterns	Medium	Better
XGB + SHAP	Global, detailed	Slow	Limited
CNN + LIME	Local, but abstract	Slow	Limited
CNN + SHAP	Complex, abstract	Very Slow	Not suitable
Hybrid RF+XGB+LIME	Local, real-time, operator-friendly	Fast	Best

stability may influence the generalizing capacity of the model. Addressing this, future research will concentrate on multi-turbine, multi-season SCADA databases including several geographical areas and grid settings. The suggested hybrid explainable framework is modular and extensible, therefore enabling quick retraining across new locations and circumstances. This guarantees that the strategy is realistic for long-term use and scalable across various wind farm settings. (See Table 5.)

5.3. Computational cost and practical feasibility

One practical concern in deploying hybrid ensemble and explainable AI models for wind turbine SCADA monitoring is the associated computational overhead. Using a workstation running an Intel Xeon processor (2.6 GHz, 32 GB RAM, no GPU acceleration), we assessed feasibility by measuring training time, model size, and explanation generation latency.

5.3.1. Model training time

Individually, Random Forest (RF) and XGBoost (XGB) took about 42 seconds and 58 seconds. The hybrid ensemble (RF + XGB) raised training time to almost 81 seconds, showing a small 1.4× overhead over a single model.

5.3.2. Model size and parameters

The RF model comprised 120 trees with a memory footprint of 45 MB, while XGB required 60 MB. The hybrid ensemble, combining both, occupied 105 MB in total. These requirements remain manageable within modern SCADA servers.

5.3.3. Explanation latency

For a single SCADA instance, SHAP took around 6.8 seconds on the same equipment, while LIME produced local explanations in about 1.2 seconds. This performance difference shows that LIME is appropriate for close to real-time diagnostic assistance where quick operator feedback is absolutely vital.

5.3.4. Scalability

Although the hybrid approach adds some computing overhead, the increased predictive accuracy and practical interpretability more than make up for it. Parallelism and batch inference can also help to lower latency in actual implementations.

6. Conclusion and future work

6.1. Conclusion

This Research produced and tested on actual real-time SCADA data a hybrid ensemble model fusing Random Forest and XGBoost to identify braking abnormalities in wind turbines. The hybrid model outperformed separate models by combining the bagging strength of Random Forest and the boosting accuracy of XGBoost to produce remarkable results with F1-score and accuracy of more than 99%. Through statistical significance testing and error measures (MAE, MSE), the model's resistance to overfitting and its great generalizing ability were further confirmed.

LIME was used to clarify the predictions of both individual and hybrid models so as to improve interpretability. The study showed that braking behavior was much influenced by temperature-related variables like Generator Temperature, Nacelle Temperature, Gearbox Oil Temperature. This knowledge helped to define thermal stress as a major cause of turbine failures, therefore emphasizing the need for better thermal maintenance techniques and cooling systems. Using LIME helped to clearly explain model decisions by pointing out important operating factors including generator temperature, voltage, and ampere levels as major influences on turbine performance. Using the Wilcoxon signed-rank test, the model's performance was verified, therefore verifying its statistical stability against similar models like Logistic Regression and XGBoost.

This solution improves the operational efficiency of wind farms, lessens downtime, and helps predictive maintenance by means of early and clear anomaly detection. Additionally directly helping towards SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 12 (Responsible Consumption and Production)

6.2. Future work

Future research can expand the hybrid model by integrating deep learning methods such as BiLSTM and CNN-GRU to more effectively capture the temporal patterns inherent in SCADA data. Furthermore, the framework can be used for multi-class fault classification, which would allow for the detection of particular fault types instead of only limiting the analysis to binary anomaly detection. Using XAI on edge computing equipment to do real-time wind turbine failure analysis is another interesting path that will help with ongoing monitoring and speedier fault resolution at the turbine level. Particularly by fusing SHAP with LIME, using several interpretability approaches might help to improve model openness and foster greater operator trust. Our study is limited to one turbine and one month of SCADA data, which may restrict generalization to seasonal patterns or other turbine types. While SMOTE addressed imbalance, long-term variability remains a challenge. Furthermore, although the hybrid ensemble improves accuracy, it may introduce higher computational costs compared to simpler models. Apart from single-turbine analysis, future research ought to look into how well the suggested system can be scaled up and how flexible it is for large-scale, multi-turbine wind farm monitoring to guarantee reliable performance in different operating environments. Ultimately, connecting the explainable findings straight with maintenance recommendations can help to enable proactive and automated fault detection, therefore enhancing the dependability and operational efficiency of wind power systems.

List of abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Network
AUC	Area Under the Curve
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CUF	Capacity Utilization Factor
DL	Deep Learning
EDA	Exploratory Data Analysis
F1	F1-Score
FP	False Positive
FN	False Negative
GB	Gradient Boosting
GRU	Gated Recurrent Unit
LIME	Local Interpretable Model-agnostic Explanations
MAE	Mean Absolute Error
ML	Machine Learning
MSE	Mean Squared Error
PR	Precision–Recall
RF	Random Forest
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
SCADA	Supervisory Control and Data Acquisition
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Oversampling Technique
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
XAI	Explainable Artificial Intelligence
XG-Boost	Extreme Gradient Boosting

CRediT authorship contribution statement

Priyadharshini Rengasamy: Writing – original draft, Resources, Formal analysis. **Rajesh R:** Supervision, Software, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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